

# Montowanie zasobów lokalnych. Zarządzanie dyskami (partycjonowanie, programowy RAID, LVM).

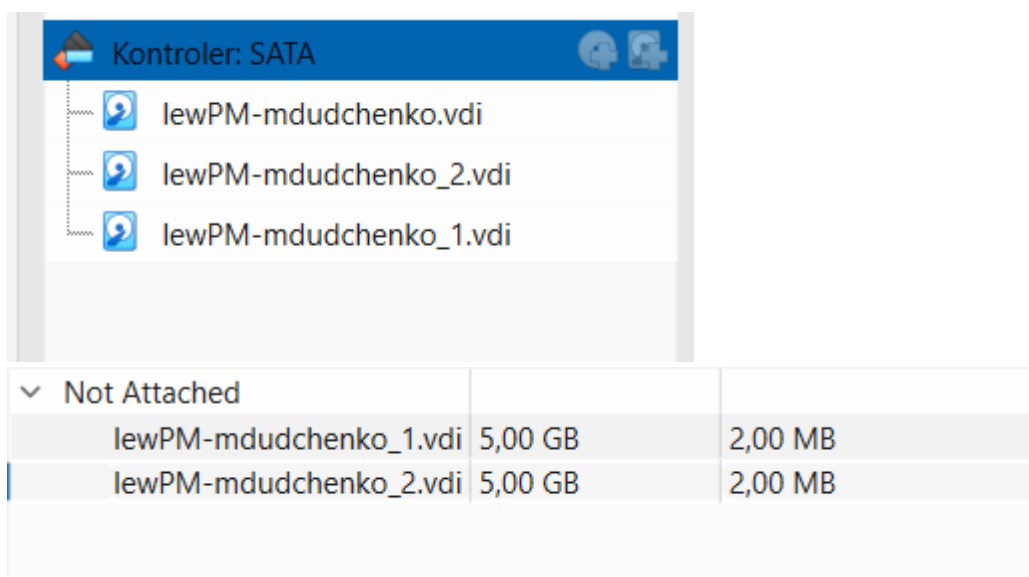
## Raport

Zadanie 1. Rozpoznaj działanie poleceń lsblk i blkid.

```
root@lew-mdudchenko:~# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0   25G  0 disk
├─sda1       8:1    0    1,9G  0 part /
├─sda2       8:2    0    954M  0 part /boot
├─sda3       8:3    0    1,9G  0 part [SWAP]
├─sda4       8:4    0      1K  0 part
├─sda5       8:5    0    8,4G  0 part /usr
├─sda6       8:6    0    2,8G  0 part /usr/local
├─sda7       8:7    0    3,7G  0 part /var
└─sda8       8:8    0    5,4G  0 part /home
sdb          8:16   0     5G  0 disk
├─sdb1       8:17   0    2,5G  0 part
└─sdb2       8:18   0    2,5G  0 part
sdc          8:32   0     5G  0 disk
├─sdc1       8:33   0    2,5G  0 part
└─sdc2       8:34   0    2,5G  0 part
sr0         11:0    1 1024M  0 rom
```

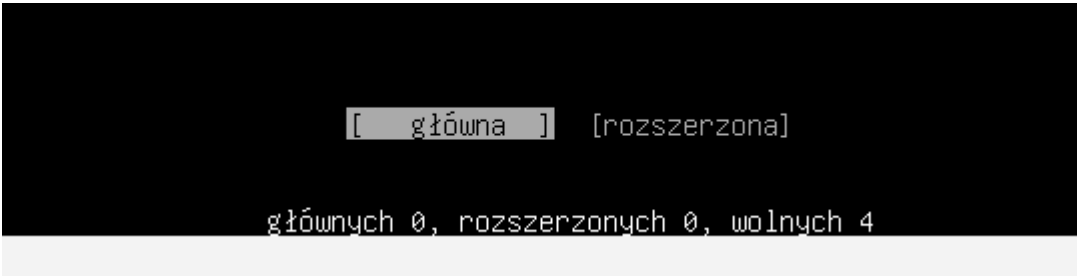
Zadanie 2.

Tworzymy maszynę wirtualną i dodajemy dyski.



robimy połowę nowych dysków

# cfdisk



| Urządzenie  | Rozruch | Początek | Koniec   | Sektory | Rozmiar | Id Typ                       |
|-------------|---------|----------|----------|---------|---------|------------------------------|
| /dev/sdc1   |         | 2048     | 5244927  | 5242880 | 2,5G    | 83 Linux                     |
| > /dev/sdc2 |         | 5244928  | 10485759 | 5240832 | 2,5G    | 82 Linux - wymiany / Solaris |



# fdisk

```

root@lew-mdudchenko:~# fdisk /dev/sdc

Witamy w programie fdisk (util-linux 2.41).
Zmiany pozostaną tylko w pamięci do chwili ich zapisania.
Przed użyciem polecenia zapisu prosimy o ostrożność.

Polecenie (m wyświetla pomoc): o
Utworzono nową etykietę dysku DOS (MBR) z identyfikatorem dysku 0xfe6f39bf.

Polecenie (m wyświetla pomoc): n
Typ partycji
  p   główna (głównych 0, rozszerzonych 0, wolnych 4)
  e   rozszerzona (kontener na partycje logiczne)
Wybór (domyślnie p): p
Numer partycji (1-4, domyślnie 1): 1
Pierwszy sektor (2048-10485759, domyślnie 2048): +2,5G
Wartość spoza zakresu.
Pierwszy sektor (2048-10485759, domyślnie 2048):
Ostatni sektor, +/-sektorów lub +/-rozmiar{K,M,G,T,P} (2048-10485759, domyślnie 10485759): +2,5G

Utworzono nową partycję 1 typu 'Linux' o rozmiarze 2,5 GiB.

Polecenie (m wyświetla pomoc): n
Typ partycji
  p   główna (głównych 1, rozszerzonych 0, wolnych 3)
  e   rozszerzona (kontener na partycje logiczne)
Wybór (domyślnie p): p
Numer partycji (2-4, domyślnie 2): 2
Pierwszy sektor (5244928-10485759, domyślnie 5244928):
Ostatni sektor, +/-sektorów lub +/-rozmiar{K,M,G,T,P} (5244928-10485759, domyślnie 10485759):

Utworzono nową partycję 2 typu 'Linux' o rozmiarze 2,5 GiB.

Polecenie (m wyświetla pomoc): t
Numer partycji (1,2, domyślnie 2): 2
Kod szesnastkowy lub alias (L wyświetla listę wszystkich): 82

Zmieniono typ partycji 'Linux' na 'Linux swap / Solaris'.

Polecenie (m wyświetla pomoc): w
Tablica partycji została zmodyfikowana.
Wywołanie ioctl() w celu ponownego odczytu tablicy partycji.
Synchronizacja dysków.

root@lew-mdudchenko:~#

```

Formatuję partycje typu Linux w systemie plików  
ext4 mount w katalogach /mnt/p1 i /mnt/p2

Dodaje do SWAP

```

root@lew-mdudchenko:~# mount /dev/sdb1 /mnt/p1
root@lew-mdudchenko:~# mount /dev/sdc1 /mnt/p2
root@lew-mdudchenko:~# mkswap /dev/sdb2
Tworzenie obszaru wymiany w wersji 1, rozmiar = 2,5 GiB (bajtów: 2683301888)
brak etykiety, UUID=2b183f05-de0c-4a2e-b86e-8d01dbcd8cb0
root@lew-mdudchenko:~# mkswap /dev/sdc2
Tworzenie obszaru wymiany w wersji 1, rozmiar = 2,5 GiB (bajtów: 2683301888)
brak etykiety, UUID=8003260a-6604-4383-90ce-37a43ccb7ae4
root@lew-mdudchenko:~# swapon /dev/sdb2
root@lew-mdudchenko:~# swapon /dev/sdc2
root@lew-mdudchenko:~# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0      0   25G  0 disk
├─sda1       8:1      0   1,9G  0 part /
├─sda2       8:2      0   954M  0 part /boot
├─sda3       8:3      0   1,9G  0 part [SWAP]
├─sda4       8:4      0     1K  0 part
├─sda5       8:5      0   8,4G  0 part /usr
├─sda6       8:6      0   2,8G  0 part /usr/local
├─sda7       8:7      0   3,7G  0 part /var
└─sda8       8:8      0   5,4G  0 part /home
sdb          8:16     0     5G  0 disk
├─sdb1       8:17     0   2,5G  0 part /mnt/p1
└─sdb2       8:18     0   2,5G  0 part [SWAP]
sdc          8:32     0     5G  0 disk
├─sdc1       8:33     0   2,5G  0 part /mnt/p2
└─sdc2       8:34     0   2,5G  0 part [SWAP]
sr0         11:0     1 1024M  0 rom
root@lew-mdudchenko:~#

```

```

root@lew-mdudchenko:~# mount /dev/sdb1 /mnt/p1
root@lew-mdudchenko:~# mount /dev/sdc1 /mnt/p2
root@lew-mdudchenko:~# mkswap /dev/sdb2
Tworzenie obszaru wymiany w wersji 1, rozmiar = 2,5 GiB (bajtów: 2683301888)
brak etykiety, UUID=2b183f05-de0c-4a2e-b86e-8d01dbcd8cb0
root@lew-mdudchenko:~# mkswap /dev/sdc2
Tworzenie obszaru wymiany w wersji 1, rozmiar = 2,5 GiB (bajtów: 2683301888)
brak etykiety, UUID=8003260a-6604-4383-90ce-37a43ccb7ae4
root@lew-mdudchenko:~# swapon /dev/sdb2
root@lew-mdudchenko:~# swapon /dev/sdc2
root@lew-mdudchenko:~# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0      0   25G  0 disk
├─sda1       8:1      0   1,9G  0 part /
├─sda2       8:2      0   954M  0 part /boot
├─sda3       8:3      0   1,9G  0 part [SWAP]
├─sda4       8:4      0     1K  0 part
├─sda5       8:5      0   8,4G  0 part /usr
├─sda6       8:6      0   2,8G  0 part /usr/local
├─sda7       8:7      0   3,7G  0 part /var
└─sda8       8:8      0   5,4G  0 part /home
sdb          8:16     0     5G  0 disk
├─sdb1       8:17     0   2,5G  0 part /mnt/p1
└─sdb2       8:18     0   2,5G  0 part [SWAP]
sdc          8:32     0     5G  0 disk
├─sdc1       8:33     0   2,5G  0 part /mnt/p2
└─sdc2       8:34     0   2,5G  0 part [SWAP]
sr0         11:0     1 1024M  0 rom
root@lew-mdudchenko:~#

```



```

root@lew-mdudchenko:~# mdadm --c
--config= --create
root@lew-mdudchenko:~# mdadm --create --verbose /dev/md11 --level=0 --raid-devices=2 /dev/sdb /dev/sdc
mdadm: chunk size defaults to 512K
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md11 started.
root@lew-mdudchenko:~# cat /proc/mdstat
-bash: cat: nie znaleziono polecenia
root@lew-mdudchenko:~# cat /proc/mdstat
Personalities : [raid0]
md11 : active raid0 sdc[1] sdb[0]
      10475520 blocks super 1.2 512k chunks

unused devices: <none>
root@lew-mdudchenko:~# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
sda          8:0    0   25G  0 disk
├─sda1       8:1    0   1,9G  0 part  /
├─sda2       8:2    0   954M  0 part  /boot
├─sda3       8:3    0   1,9G  0 part  [SWAP]
├─sda4       8:4    0    1K  0 part
├─sda5       8:5    0   8,4G  0 part  /usr
├─sda6       8:6    0   2,8G  0 part  /usr/local
├─sda7       8:7    0   3,7G  0 part  /var
└─sda8       8:8    0   5,4G  0 part  /home
sdb          8:16   0    5G  0 disk
└─md11       9:11   0   10G  0 raid0
sdc          8:32   0    5G  0 disk
└─md11       9:11   0   10G  0 raid0
sdd          8:48   0    5G  0 disk
sde          8:64   0    5G  0 disk
sdf          8:80   0    5G  0 disk
sdg          8:96   0    5G  0 disk
sdh          8:112  0    5G  0 disk
sdi          8:128  0    5G  0 disk
sdj          8:144  0    5G  0 disk
sdk          8:160  0    5G  0 disk
sr0         11:0    1 1024M  0 rom
root@lew-mdudchenko:~#

```

```

root@lew-mdudchenko:~# mdadm --create --verbose /dev/md12 --level=0 --raid-devices=2 /dev/sdd /dev/sde
mdadm: chunk size defaults to 512K
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md12 started.
root@lew-mdudchenko:~# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
sda          8:0    0   25G  0 disk
├─sda1       8:1    0   1,9G  0 part  /
├─sda2       8:2    0   954M  0 part  /boot
├─sda3       8:3    0   1,9G  0 part  [SWAP]
├─sda4       8:4    0    1K  0 part
├─sda5       8:5    0   8,4G  0 part  /usr
├─sda6       8:6    0   2,8G  0 part  /usr/local
├─sda7       8:7    0   3,7G  0 part  /var
└─sda8       8:8    0   5,4G  0 part  /home
sdb          8:16   0    5G  0 disk
└─md11       9:11   0   10G  0 raid0
sdc          8:32   0    5G  0 disk
└─md11       9:11   0   10G  0 raid0
sdd          8:48   0    5G  0 disk
└─md12       9:12   0   10G  0 raid0
sde          8:64   0    5G  0 disk
└─md12       9:12   0   10G  0 raid0
sdf          8:80   0    5G  0 disk
sdg          8:96   0    5G  0 disk
sdh          8:112  0    5G  0 disk
sdi          8:128  0    5G  0 disk
sdj          8:144  0    5G  0 disk
sdk          8:160  0    5G  0 disk
sr0         11:0    1 1024M  0 rom
root@lew-mdudchenko:~#

```

## Raid 1

```

root@lew-mdudchenko:~# mdadm --create --verbose /dev/md1 --level=1 --raid-devices=2 /dev/md11 /dev/md12
To optimize recovery speed, it is recommended to enable write-indent bitmap, do you want to enable it now? [y/N]? y
mdadm: Note: this array has metadata at the start and
may not be suitable as a boot device. If you plan to
store '/boot' on this device please ensure that
your boot-loader understands md/v1.x metadata, or use
--metadata=0.90
mdadm: size set to 10466304K
Continue creating array [y/N]? y
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md1 started.
root@lew-mdudchenko:~# lsblk

```

| NAME   | MAJ:MIN | RM | SIZE  | RO | TYPE  | MOUNTPOINTS |
|--------|---------|----|-------|----|-------|-------------|
| sda    | 8:0     | 0  | 25G   | 0  | disk  |             |
| └─sda1 | 8:1     | 0  | 1,9G  | 0  | part  | /           |
| └─sda2 | 8:2     | 0  | 954M  | 0  | part  | /boot       |
| └─sda3 | 8:3     | 0  | 1,9G  | 0  | part  | [SWAP]      |
| └─sda4 | 8:4     | 0  | 1K    | 0  | part  |             |
| └─sda5 | 8:5     | 0  | 8,4G  | 0  | part  | /usr        |
| └─sda6 | 8:6     | 0  | 2,8G  | 0  | part  | /usr/local  |
| └─sda7 | 8:7     | 0  | 3,7G  | 0  | part  | /var        |
| └─sda8 | 8:8     | 0  | 5,4G  | 0  | part  | /home       |
| sdb    | 8:16    | 0  | 5G    | 0  | disk  |             |
| └─md11 | 9:11    | 0  | 10G   | 0  | raid0 |             |
| └─md1  | 9:1     | 0  | 10G   | 0  | raid1 |             |
| sdc    | 8:32    | 0  | 5G    | 0  | disk  |             |
| └─md11 | 9:11    | 0  | 10G   | 0  | raid0 |             |
| └─md1  | 9:1     | 0  | 10G   | 0  | raid1 |             |
| sdd    | 8:48    | 0  | 5G    | 0  | disk  |             |
| └─md12 | 9:12    | 0  | 10G   | 0  | raid0 |             |
| └─md1  | 9:1     | 0  | 10G   | 0  | raid1 |             |
| sde    | 8:64    | 0  | 5G    | 0  | disk  |             |
| └─md12 | 9:12    | 0  | 10G   | 0  | raid0 |             |
| └─md1  | 9:1     | 0  | 10G   | 0  | raid1 |             |
| sdf    | 8:80    | 0  | 5G    | 0  | disk  |             |
| sdg    | 8:96    | 0  | 5G    | 0  | disk  |             |
| sdh    | 8:112   | 0  | 5G    | 0  | disk  |             |
| sdj    | 8:128   | 0  | 5G    | 0  | disk  |             |
| sdj    | 8:144   | 0  | 5G    | 0  | disk  |             |
| sdj    | 8:160   | 0  | 5G    | 0  | disk  |             |
| sr0    | 11:0    | 1  | 1024M | 0  | rom   |             |

```

root@lew-mdudchenko:~#

```

## Raid 5

```

root@lew-mdudchenko:~# mdadm --create --verbose /dev/md100 --level=5 --raid-devices=4 /dev/sdf /dev/sdg /dev/sdh /dev/sdi
To optimize recovery speed, it is recommended to enable write-indent bitmap, do you want to enable it now? [y/N]? y
mdadm: layout defaults to left-symmetric
mdadm: layout defaults to left-symmetric
mdadm: chunk size defaults to 512K
mdadm: size set to 5237760K
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md100 started.
root@lew-mdudchenko:~# cat /proc/mdstat
Personalities : [raid0] [raid1] [raid6] [raid5] [raid4]
md100 : active raid5 sdi[4] sdh[2] sdg[1] sdf[0]
15713280 blocks super 1.2 level 5, 512k chunk, algorithm 2 [4/3] [UUU_]
[>.....] recovery = 4.5% (239160/5237760) finish=1.7min speed=47832K/sec
bitmap: 1/1 pages [4KB], 65536KB chunk

md1 : active raid1 md12[1] md11[0]
10466304 blocks super 1.2 [2/2] [UU]
bitmap: 0/1 pages [0KB], 65536KB chunk

md12 : active raid0 sde[1] sdd[0]
10475520 blocks super 1.2 512k chunks

md11 : active raid0 sdc[1] sdb[0]
10475520 blocks super 1.2 512k chunks

unused devices: <none>
root@lew-mdudchenko:~#

```

## Formatuje do ext4 md1 i md100

```
root@lew-mdudchenko:~# mkfs.ext4 /dev/md1
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 2616576 4k blocks and 655360 inodes
Filesystem UUID: 5c6a77a0-cefa-4d6e-bb2a-a56b124b8534
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

root@lew-mdudchenko:~# mkfs.ext4 /dev/md100
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 3928320 4k blocks and 983040 inodes
Filesystem UUID: 5d5c58da-145f-43d0-9107-28fd0338fc3b
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

root@lew-mdudchenko:~#
```

Ilość dostępnego miejsca

```

sda1      8:1      0  1,9G  0 part  /
sda2      8:2      0  954M  0 part  /boot
sda3      8:3      0  1,9G  0 part  [SWAP]
sda4      8:4      0    1K  0 part
sda5      8:5      0  8,4G  0 part  /usr
sda6      8:6      0  2,8G  0 part  /usr/local
sda7      8:7      0  3,7G  0 part  /var
sda8      8:8      0  5,4G  0 part  /home
sdb        8:16     0    5G  0 disk
└─md11     9:11     0   10G  0 raid0
   └─md1    9:1      0   10G  0 raid1 /raid/mirror
sdc        8:32     0    5G  0 disk
└─md11     9:11     0   10G  0 raid0
   └─md1    9:1      0   10G  0 raid1 /raid/mirror
sdd        8:48     0    5G  0 disk
└─md12     9:12     0   10G  0 raid0
   └─md1    9:1      0   10G  0 raid1 /raid/mirror
sde        8:64     0    5G  0 disk
└─md12     9:12     0   10G  0 raid0
   └─md1    9:1      0   10G  0 raid1 /raid/mirror
sdf        8:80     0    5G  0 disk
└─md100    9:100    0   15G  0 raid5 /raid/raid5
sdg        8:96     0    5G  0 disk
└─md100    9:100    0   15G  0 raid5 /raid/raid5
sdh        8:112    0    5G  0 disk
└─md100    9:100    0   15G  0 raid5 /raid/raid5
sdi        8:128    0    5G  0 disk
└─md100    9:100    0   15G  0 raid5 /raid/raid5
sdj        8:144    0    5G  0 disk
sdk        8:160    0    5G  0 disk
sr0       11:0      1 1024M  0 rom
root@lew-mdudchenko:~# df -h
System plików rozmi. użyte dost. %uż. zamont. na
udev          960M      0  960M   0% /dev
tmpfs         198M   656K  197M   1% /run
/dev/sda1     1,8G   4,8M   1,7G   1% /
/dev/sda5     8,2G   802M   7,0G  11% /usr
tmpfs         987M      0  987M   0% /dev/shm
tmpfs         5,0M      0   5,0M   0% /run/lock
tmpfs         1,0M      0   1,0M   0% /run/credentials/systemd-journald.service
tmpfs         987M      0  987M   0% /tmp
/dev/sda2     921M  121M  737M  15% /boot
/dev/sda7     3,6G   333M   3,1G  10% /var
/dev/sda6     2,7G   788K   2,6G   1% /usr/local
/dev/sda8     5,3G   1,5M   5,0G   1% /home
tmpfs         1,0M      0   1,0M   0% /run/credentials/getty@tty1.service
tmpfs         198M   4,0K  198M   1% /run/user/0
/dev/md1      9,8G   2,1M   9,3G   1% /raid/mirror
/dev/md100    15G   2,1M   14G   1% /raid/raid5
root@lew-mdudchenko:~#

```

sprzątam wszystko

```

root@lew-mdudchenko:~# mdadm --stop /dev/md1
mdadm: stopped /dev/md1
root@lew-mdudchenko:~# umount /dev/md1
umount: /dev/md1: nie podano punktu montowania.
root@lew-mdudchenko:~# mdadm --stop /dev/md11
mdadm: stopped /dev/md11
root@lew-mdudchenko:~# mdadm --stop /dev/md12
mdadm: stopped /dev/md12
root@lew-mdudchenko:~# mdadm --stop /dev/md100
mdadm: Cannot get exclusive access to /dev/md100:Perhaps a running process, mounted filesystem or active volume group?
root@lew-mdudchenko:~# umount /dev/md100
root@lew-mdudchenko:~# mdadm --stop /dev/md100
mdadm: stopped /dev/md100
root@lew-mdudchenko:~# cat /proc/mdstat
Personalities : [raid0] [raid1] [raid6] [raid5] [raid4]
unused devices: <none>
root@lew-mdudchenko:~# _

```

Tworzę nowa maszynie i pobieram pakiet lvm2.  
nowe dyski systemowe

```

Wybieranie wcześniej niewybranego pakietu liblvm2cmd2.03:amd64.
Przygotowywanie do rozpakowania pakietu .../2-liblvm2cmd2.03_2.03.31-2_amd64.deb ...
Rozpakowywanie pakietu liblvm2cmd2.03:amd64 (2.03.31-2) ...
Wybieranie wcześniej niewybranego pakietu dmeventd.
Przygotowywanie do rozpakowania pakietu .../3-dmeventd_2%3a1.02.205-2_amd64.deb ...
Rozpakowywanie pakietu dmeventd (2:1.02.205-2) ...
Wybieranie wcześniej niewybranego pakietu lvm2.
Przygotowywanie do rozpakowania pakietu .../4-lvm2_2.03.31-2_amd64.deb ...
Rozpakowywanie pakietu lvm2 (2.03.31-2) ...
Wybieranie wcześniej niewybranego pakietu thin-provisioning-tools.
Przygotowywanie do rozpakowania pakietu .../5-thin-provisioning-tools_1.1.0-4+b1_amd64.deb ...
Rozpakowywanie pakietu thin-provisioning-tools (1.1.0-4+b1) ...
Konfigurowanie pakietu libdevmapper-event1.02.1:amd64 (2:1.02.205-2) ...
Konfigurowanie pakietu thin-provisioning-tools (1.1.0-4+b1) ...
Konfigurowanie pakietu libaio1t64:amd64 (0.3.113-8+b1) ...
Konfigurowanie pakietu liblvm2cmd2.03:amd64 (2.03.31-2) ...
Konfigurowanie pakietu dmeventd (2:1.02.205-2) ...
Created symlink '/etc/systemd/system/sockets.target.wants/dm-event.socket' → '/usr/lib/systemd/system/dm-event.socket'.
dm-event.service is a disabled or a static unit, not starting it.
Konfigurowanie pakietu lvm2 (2.03.31-2) ...
Created symlink '/etc/systemd/system/sysinit.target.wants/blk-availability.service' → '/usr/lib/systemd/system/blk-availability.service'.
Created symlink '/etc/systemd/system/sysinit.target.wants/lvm2-monitor.service' → '/usr/lib/systemd/system/lvm2-monitor.service'.
Created symlink '/etc/systemd/system/sysinit.target.wants/lvm2-lvmpolld.socket' → '/usr/lib/systemd/system/lvm2-lvmpolld.socket'.
Przetwarzanie wyzwalaczy pakietu initramfs-tools (0.148.3)...
update-initramfs: Generating /boot/initrd.img-6.12.74+deb13+1-amd64
Przetwarzanie wyzwalaczy pakietu libc-bin (2.41-12+deb13u2)...
Przetwarzanie wyzwalaczy pakietu man-db (2.13.1-1)...
root@lew-mdudchenko:~# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda         8:0    0   25G  0 disk
├─sda1      8:1    0   1,9G  0 part /
├─sda2      8:2    0   954M  0 part /boot
├─sda3      8:3    0   1,9G  0 part [SWAP]
├─sda4      8:4    0    1K  0 part
├─sda5      8:5    0    8,4G  0 part /usr
├─sda6      8:6    0    2,8G  0 part /usr/local
├─sda7      8:7    0    3,7G  0 part /var
└─sda8      8:8    0    5,4G  0 part /home
sdb         8:16   0    5G  0 disk
sdc         8:32   0    5G  0 disk
sdd         8:48   0    5G  0 disk
sde         8:64   0    5G  0 disk
sdf         8:80   0    5G  0 disk
sdg         8:96   0    5G  0 disk
sdh         8:112  0    5G  0 disk
sdi         8:128  0    5G  0 disk
sdj         8:144  0    5G  0 disk
sdk         8:160  0    5G  0 disk
sr0        11:0    1 1024M  0 rom
root@lew-mdudchenko:~#

```

Tworzę woluminy fizyczne z nowych dysków  
Tworzę z grupę woluminów vgów z dostępnych  
woluminów fizycznych

```

root@lew-mdudchenko:~# pvcreate /dev/sdk
Physical volume "/dev/sdk" successfully created.
root@lew-mdudchenko:~# lvm diskscan
/dev/sda1 [      1,86 GiB]
/dev/sdh [      5,00 GiB] LVM physical volume
/dev/sdi [      5,00 GiB] LVM physical volume
/dev/sdj [      5,00 GiB] LVM physical volume
/dev/sdb [      5,00 GiB] LVM physical volume
/dev/sdk [      5,00 GiB] LVM physical volume
/dev/sda2 [    954,00 MiB]
/dev/sda3 [      1,86 GiB]
/dev/sdc [      5,00 GiB] LVM physical volume
/dev/sdd [      5,00 GiB] LVM physical volume
/dev/sda5 [      8,38 GiB]
/dev/sda6 [      2,79 GiB]
/dev/sde [      5,00 GiB] LVM physical volume
/dev/sda7 [      3,72 GiB]
/dev/sda8 [      5,44 GiB]
/dev/sdf [      5,00 GiB] LVM physical volume
/dev/sdg [      5,00 GiB] LVM physical volume
0 disks
7 partitions
10 LVM physical volume whole disks
0 LVM physical volumes

```

```

root@lew-mdudchenko:~# vgcreate vglew /dev/sdb /dev/sdd /dev/sdc /dev/sde /dev/sdf /dev/sdg /dev/sdh /dev/sdi /dev/sdj /dev/sdk
Volume group "vglew" successfully created

```

```

root@lew-mdudchenko:~# vgcreate vglew /dev/sdb /dev/sdd /dev/sdc /dev/sde /dev/sdf /dev/sdg /dev/sdh /dev/sdi /dev/sdj /dev/sdk
Volume group "vglew" successfully created

```

```

root@lew-mdudchenko:~# lvm diskscan
/dev/sda1 [      1,86 GiB]
/dev/sdh [      5,00 GiB] LVM physical volume
/dev/sdi [      5,00 GiB] LVM physical volume
/dev/sdj [      5,00 GiB] LVM physical volume
/dev/sdb [      5,00 GiB] LVM physical volume
/dev/sdk [      5,00 GiB] LVM physical volume
/dev/sda2 [    954,00 MiB]
/dev/sda3 [      1,86 GiB]
/dev/sdc [      5,00 GiB] LVM physical volume
/dev/sdd [      5,00 GiB] LVM physical volume
/dev/sda5 [      8,38 GiB]
/dev/sda6 [      2,79 GiB]
/dev/sde [      5,00 GiB] LVM physical volume
/dev/sda7 [      3,72 GiB]
/dev/sda8 [      5,44 GiB]
/dev/sdf [      5,00 GiB] LVM physical volume
/dev/sdg [      5,00 GiB] LVM physical volume
0 disks
7 partitions
10 LVM physical volume whole disks
0 LVM physical volumes

```

```

root@lew-mdudchenko:~# vgdisplay
--- Volume group ---
VG Name                vglew
System ID
Format                 lvm2
Metadata Areas         10
Metadata Sequence No   1
VG Access              read/write
VG Status              resizable
MAX LV                 0
Cur LV                0
Open LV                0
Max PV                 0
Cur PV                10
Act PV                 10
VG Size                49,96 GiB
PE Size                4,00 MiB
Total PE               12790
Alloc PE / Size        0 / 0
Free PE / Size         12790 / 49,96 GiB
VG UUID                jXnkRg-1rTJ-byQ3-5ZMe-nUKM-0JYU-tvSImr

```

```

root@lew-mdudchenko:~# _

```

```
oot@lew-mdudchenko:~# lvcreate -L 2G vglew -n data
Logical volume "data" created.
oot@lew-mdudchenko:~# lvcreate -L 5G vglew -n backup
Logical volume "backup" created.
oot@lew-mdudchenko:~# lvcreate -L 3G vglew -n test
Logical volume "test" created.
oot@lew-mdudchenko:~# vgdisplay
--- Volume group ---
VG Name                vglew
System ID
Format                 lvm2
Metadata Areas         10
Metadata Sequence No   4
VG Access              read/write
VG Status              resizable
MAX LV                 0
Cur LV                3
Open LV                0
Max PV                 0
Cur PV                10
Act PV                10
VG Size                49,96 GiB
PE Size                4,00 MiB
Total PE               12790
Alloc PE / Size        2560 / 10,00 GiB
Free PE / Size         10230 / 39,96 GiB
VG UUID                jXnkRg-1rTJ-byQ3-5ZMe-nUKM-0JYU-tVSImp

oot@lew-mdudchenko:~#
```

## Formatuję resztę

```
root@lew-mdudchenko:~# mkfs.ext4 /dev/vglew/data
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 524288 4k blocks and 131072 inodes
Filesystem UUID: 236a4789-614c-44ec-838b-cc8527809b1e
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

root@lew-mdudchenko:~# mkfs.ext4 /dev/vglew/backup
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: 5d941fbf-a14a-43fc-a2ef-8a55f8366965
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

root@lew-mdudchenko:~# mkfs.ext4 /dev/vglew/test
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 786432 4k blocks and 196608 inodes
Filesystem UUID: 50b62219-b02d-429f-bdbc-a5befbac2fa7
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

root@lew-mdudchenko:~#
```

```

root@lew-mdudchenko:~# mount /dev/vglew/data /var/data
mount: /var/data: punkt montowania nie istnieje.
        dmesg(1) może podać więcej informacji o nieudanym wywołaniu systemowym mount.
root@lew-mdudchenko:~# mkdir -p /var/data /var/test /var/backup
root@lew-mdudchenko:~# mount /dev/vglew/data /var/data
root@lew-mdudchenko:~# mount /dev/vglew/backup /var/backup
root@lew-mdudchenko:~# mount /dev/vglew/test /var/test/
root@lew-mdudchenko:~# ls -l /var/
razem 64
drwxr-xr-x  3 root root  4096 04-21 18:01 backup
drwxr-xr-x  2 root root  4096 04-21 17:49 backups
drwxr-xr-x 10 root root  4096 04-14 16:05 cache
drwxr-xr-x  3 root root  4096 04-21 18:01 data
drwxr-xr-x 22 root root  4096 04-14 16:05 lib
drwxr-xr-x  2 root root  4096 03-02 22:50 local
lrwxrwxrwx  1 root root    9 04-14 15:58 lock -> /run/lock
drwxr-xr-x  7 root root  4096 04-21 17:36 log
drwx----- 2 root root 16384 04-14 15:57 lost+found
drwxrwsr-x  2 root mail  4096 04-14 15:58 mail
drwxr-xr-x  2 root root  4096 04-14 15:58 opt
lrwxrwxrwx  1 root root    4 04-14 15:58 run -> /run
drwxr-xr-x  3 root root  4096 04-14 15:58 spool
drwxr-xr-x  3 root root  4096 04-21 18:01 test
drwxrwxrwt  3 root root  4096 04-21 17:37 tmp
root@lew-mdudchenko:~#

```

```

root@lew-mdudchenko:~# lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                                  8:0      0   25G  0 disk
├─sda1                              8:1      0   1,9G  0 part /
├─sda2                              8:2      0   954M  0 part /boot
├─sda3                              8:3      0   1,9G  0 part [SWAP]
├─sda4                              8:4      0     1K  0 part
├─sda5                              8:5      0   8,4G  0 part /usr
├─sda6                              8:6      0   2,8G  0 part /usr/local
├─sda7                              8:7      0   3,7G  0 part /var
└─sda8                              8:8      0   5,4G  0 part /home
sdb                                  8:16     0     5G  0 disk
├─vglew-data                        254:0     0     2G  0 lvm  /var/data
└─vglew-backup                      254:1     0     5G  0 lvm  /var/backup
sdc                                  8:32     0     5G  0 disk
├─vglew-test                        254:2     0     3G  0 lvm  /var/test
sdd                                  8:48     0     5G  0 disk
├─vglew-backup                      254:1     0     5G  0 lvm  /var/backup
sde                                  8:64     0     5G  0 disk
sdf                                  8:80     0     5G  0 disk
sdg                                  8:96     0     5G  0 disk
sdh                                  8:112    0     5G  0 disk
sdi                                  8:128    0     5G  0 disk
sdj                                  8:144    0     5G  0 disk
sdk                                  8:160    0     5G  0 disk
sr0                                  11:0     1 1024M  0 rom
root@lew-mdudchenko:~#

```

dodaje 3 nowe dyski

```
root@lew-mdudchenko:~# vgextend vglew /dev/sdl /dev/sdn /dev/sdm
Physical volume "/dev/sdl" successfully created.
Physical volume "/dev/sdn" successfully created.
Physical volume "/dev/sdm" successfully created.
Volume group "vglew" successfully extended
root@lew-mdudchenko:~# lsblk
```

```
--- Logical volume ---
LV Path                /dev/vglew/mylogs
LV Name                 mylogs
VG Name                 vglew
LV UUID                 6o2Jzf-4jlf-LtVC-hnCK-HDQX-OMCB-ZDbvDn
LV Write Access         read/write
LV Creation host, time  lew-mdudchenko, 2026-04-21 18:38:07 +0200
LV Status                available
# open                   0
LV Size                  7,00 GiB
Current LE               1792
Segments                1
Allocation               inherit
Read ahead sectors      auto
 - currently set to     256
Block device             254:3
```

## Montuje

```
root@lew-mdudchenko:~# mkdir -p /var/mylogs
root@lew-mdudchenko:~# mkfs.ext4 /dev/vglew/mylogs
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 1835008 4k blocks and 458752 inodes
Filesystem UUID: 6beb01ea-3b2c-462b-bb09-360ed3aa9bf6
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

root@lew-mdudchenko:~# mount /dev/vglew/mylogs /var/mylogs/
root@lew-mdudchenko:~# _
```

```

root@lew-mdudchenko:~# lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                  8:0    0   25G  0 disk
├─sda1               8:1    0   1,9G  0 part /
├─sda2               8:2    0   954M  0 part /boot
├─sda3               8:3    0   1,9G  0 part [SWAP]
├─sda4               8:4    0     1K  0 part
├─sda5               8:5    0   8,4G  0 part /usr
├─sda6               8:6    0   2,8G  0 part /usr/local
├─sda7               8:7    0   3,7G  0 part /var
└─sda8               8:8    0   5,4G  0 part /home
sdb                  8:16    0    5G  0 disk
├─vglew-data         254:0    0    2G  0 lvm
└─vglew-backup       254:1    0    5G  0 lvm
sdc                  8:32    0    5G  0 disk
└─vglew-test         254:2    0    3G  0 lvm
sdd                  8:48    0    5G  0 disk
└─vglew-backup       254:1    0    5G  0 lvm
sde                  8:64    0    5G  0 disk
sdf                  8:80    0    5G  0 disk
sdg                  8:96    0    5G  0 disk
sdh                  8:112   0    5G  0 disk
sdi                  8:128   0    5G  0 disk
sdj                  8:144   0    5G  0 disk
sdk                  8:160   0    5G  0 disk
sdl                  8:176   0   10G  0 disk
└─vglew-mylogs       254:3    0    7G  0 lvm /var/mylogs
sdm                  8:192   0   10G  0 disk
sdn                  8:208   0   10G  0 disk
sr0                  11:0    1 1024M  0 rom
root@lew-mdudchenko:~#

```

## Usuwamy

```

root@lew-mdudchenko:~# lvremove /dev/vglew/test _

```

```
LV Path          /dev/vg1ew/data
LV Name          data
VG Name          vg1ew
LV UUID          ECgkji-VVC7-ckNc-6hSk-b89g-4dxC-c2igLy
LV Write Access  read/write
LV Creation host, time lew-mdudchenko, 2026-04-21 17:59:34 +0200
LV Status        available
# open           0
LV Size          2,00 GiB
Current LE       512
Segments         1
Allocation       inherit
Read ahead sectors auto
- currently set to 256
Block device     254:0
```

--- Logical volume ---

```
LV Path          /dev/vg1ew/backup
LV Name          backup
VG Name          vg1ew
LV UUID          dvTWGk-2ytl-2xXK-Q0rd-FGxT-3eox-uG1W1B
LV Write Access  read/write
LV Creation host, time lew-mdudchenko, 2026-04-21 17:59:49 +0200
LV Status        available
# open           0
LV Size          5,00 GiB
Current LE       1280
Segments         2
Allocation       inherit
Read ahead sectors auto
- currently set to 256
Block device     254:1
```

--- Logical volume ---

```
LV Path          /dev/vg1ew/mylogs
LV Name          mylogs
VG Name          vg1ew
LV UUID          6o2Jzf-4jlf-LtVC-hnCK-HDQX-OMCB-ZDbvDn
LV Write Access  read/write
LV Creation host, time lew-mdudchenko, 2026-04-21 18:38:07 +0200
LV Status        available
# open           1
LV Size          7,00 GiB
Current LE       1792
Segments         1
Allocation       inherit
Read ahead sectors auto
- currently set to 256
Block device     254:3
```